

DIANE OXALES

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EDUCATION

Carleton University – Ottawa, ON

Sep. 2023 – April. 2028 (expected)

- Bachelor of Computer Science Honours, Stream: Software Engineering, Co-op
- Third Year Standing 3.61/4 (A-)
- Availability: Summer Term (May – August) 2026

SKILLS SUMMARY

Technical

- **Networking & Systems:** OSI Model, TCP/IP, Network troubleshooting
- **Programming:** C, C++, JavaScript, Python, Java, HTML/CSS
- **Tools:** Git, Oracle VM, REST API, Microsoft Office Suite, Google Tools, VS Code, IntelliJ, Linux Environments

Professional

- Problem-solving and systematic troubleshooting.
- Clear written and verbal technical communication.
- Quick learner; effective in team-based environments.

RELEVANT WORK EXPERIENCE

Web Applications Teaching Assistant – Carleton University

Present

- Support labs and tutorials by explaining **client-server interactions**, form validation, and basic **HTTP request-response workflows**.
- Provide technical feedback on assignments, focusing on **code correctness, readability, and best practices** in front-end development.
- Guide students in troubleshooting **cross-browser compatibility**, CSS specificity, and responsive design issues.

STEM Outreach Instructor – Virtual Ventures

September. 2025 - December. 2025

- Deliver hands-on workshops introducing **computing, networking, and engineering fundamentals** to diverse audiences.
- Explain complex STEM concepts in a clear, structured communication, similar to customer-facing technical support.
- Adapt explanations in real time based on user feedback to resolve misunderstandings efficiently.
- Create lessons STEM-related to be taught to youth (kindergarten- Grade 12), reinforcing analytical thinking.

Tutoring

March. 2025 – July. 2025

- **Teach Mathematics** (Grade 1-12), and Chemistry (Grade 11-12) to strengthen the **understanding** of students in areas of struggle.
- Provide practice problems and advice to meet the goals of the student.
- Collaborate with parents to align teaching strategies with academic goals, ensuring transparent communication and success.

PROJECTS

CUInSpace Flight Simulator — C

- Implemented **dynamic memory** management to use memory only when necessary.
- Ensured **concurrency** in the program using threads & semaphores.
- Worked with **linked lists** for easy insertion & deletion when provided large amounts of data.

Auto Park Graphical User Interface (GUI) — Java

- Developed a graphical user interface using the **Model, View, Controller (MVC)** paradigm in Java.
- Implemented different access modifiers to ensure security from outside users.

Full-Stack Bible Application — JavaScript / Node.js / REST API

- Designed and developed a **frontend and backend web application** using JavaScript and RESTful APIs.
- Implemented **client-server communication**, reinforcing TCP/IP and request-response concepts.
- Built backend services to handle user requests, data persistence, and error handling.

VOLUNTEER

Tech Booth Controller

March. 2018 – June. 2023

- Created and managed slideshow presentations for Weekly Masses to engage the Parishioners.
- Resolved technical issues in real time to ensure consistent audio quality for both choir and clergy.